EXACT QUOTIENT-MULTIPLICITY EXCLUSION OF 18-ELEMENT EQUALITY CANDIDATES IN THOMPSON'S GROUP F

ARGUS OPEN RESEARCH PROJECT

Abstract. In the investigation of finite Følner/amenability equality candidates in Thompson's group F on the ring-shifted $p = 22$ template, we prove that all 18-element equality candidates are impossible: the ordered-grid balance requires adjacent rows to share at least 15 right-quotient products, whereas exhaustive exact evaluation of the quotient table proves that the maximum negative-quotient multiplicity is 8.

1. Main Exclusion Theorem

Theorem 1.1. *In the $p = 22$ two-column derived template inside Thompson's group F, any 18-element configuration satisfying the boundary-cancellation equality between generators x_0 and x_1 requires adjacent row fibres to share at least 15 common right quotients, whereas the complete exact quotient table has maximum off-diagonal quotient multiplicity $8 < 15$. Hence no 18-element equality witness exists in this family.*

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